Team Check-In 2: Prototyping

MIME 497 Group 08B

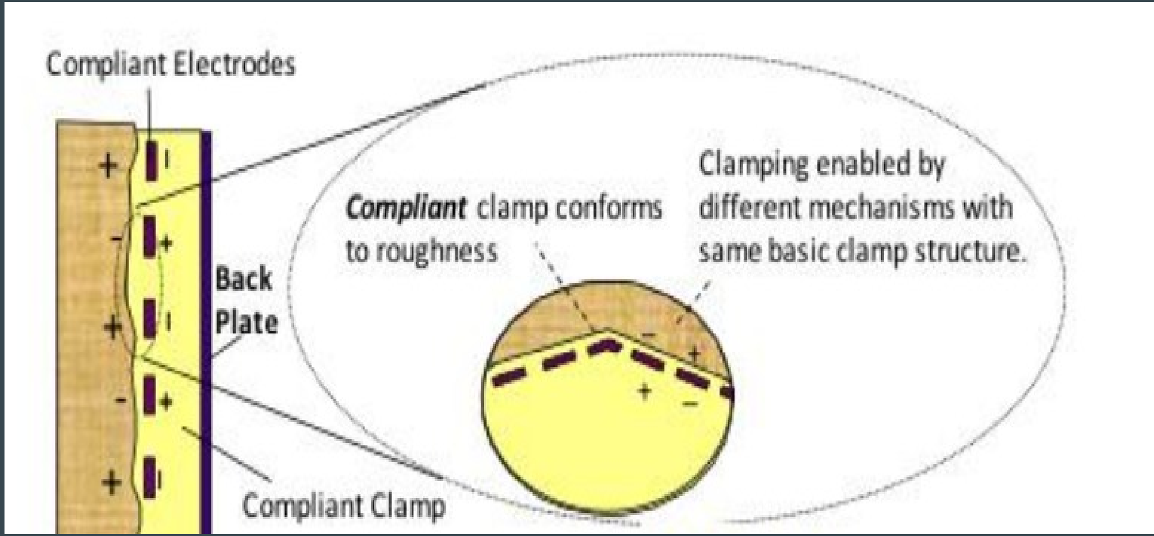
Synopsis of Project

- Electrostatic Adhesion with explosive welding anchoring subsystem
- Folding legs with flexible sections for uneven terrain
- Magnets = Thermite charges
- Toothpicks = Electrodes that conduct electricity
- Cloth = Gripper pads that induce electrostatic force

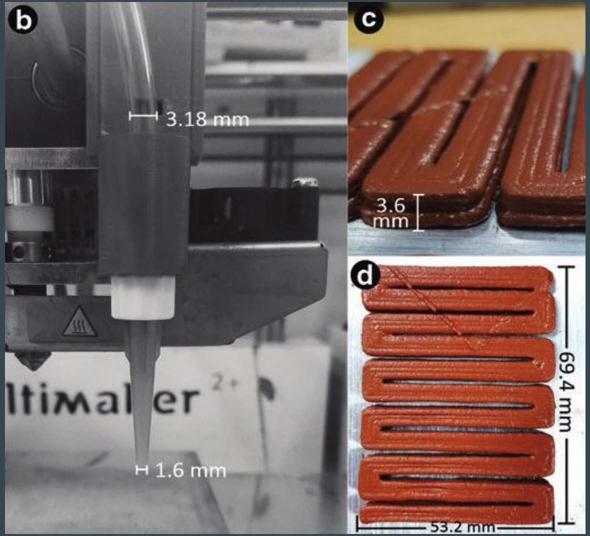
Prototype anchoring in worst case scenario

Proof of Concept Examples

- The electrostatic grippers are capable of creating a soft landing on the surface
- Flexible design allows our feet to somewhat mould to the shape of the surface
- Explosive welding accounts for redundancy within the electroadhesion

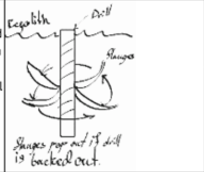
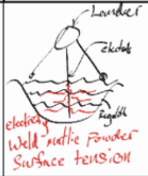
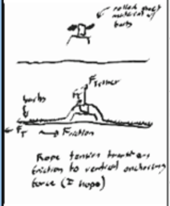
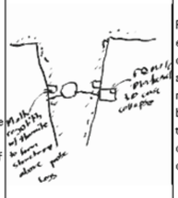
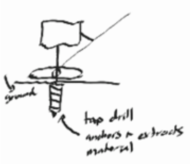
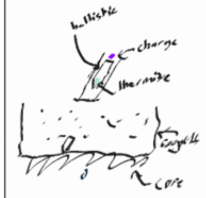
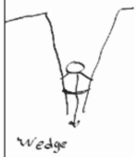
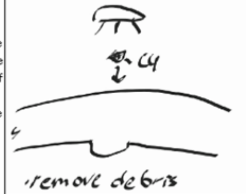
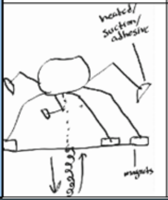
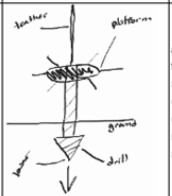
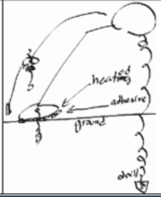


Electrostatic gripper pads with electrodes illustration

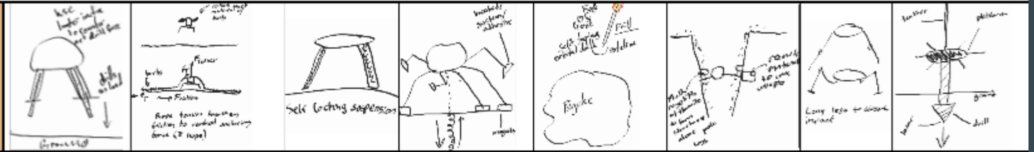


3D-printed thermite paste

Morph Matrix

Morph-Matrix								
Project:	**Psyche Space Elevator Attachment**							
	Concept	***Discriptions***	***Concept***	***Discriptions***	***Concept***	***Discriptions***	***Concept***	***Discriptions***
Astroid Surface Attachment Method		During landing the feet of the lander drill in using the inertia of the lander. Potential applications with the flanged drill head concept and self locking suspension		Self-Tapping Metal Rod launched from orbit to impact into regolith with anchor loops at the end to attach bottom of tether. Potential applications with flange drill head, Wedge and Platform Cocepts		Drill has counter-rotational flanges that extend out to dig into the material locking it into place as upward force is applied and-or the drill is backed out.		Lander uses an electrode forced into the mostly metal regolith using a supercapacitor connected to the electrode will discharge electrical potential to heat up and melt the metal, welding the lander to the surface. Potential use of Flange drill to increase the surface area of the electrode
		During landing, sheets of material are shot outwards from the lander. The goal is to maximize surface area contact with the surface of Psyche to create a large amount of friction. This friction is further increased by barbs and material choices. Hopefully, the rope like/elastic nature of the material cuases tension to be diverted to convert the upward pull of the tether into a horizontal pull that can be countered by friction.		Remove material form an existing steep crater to cause a collapse that holds the anchor and craft down. Alternatively, the regolith and/or core metal could be melted above the crafts legs to hold it in place. This melting could be done with thermite or other welding methods.		This is a spin on a drill based approach, where the conventional drill bit is replaced with a special bit that is similar to a screw tap. This would would mean that the drill would pull itself deeper upon initially entering the material, and there is potential for it to remove material to the surface by spinning in the opposite direction (in the same way that a screw tap turns 4 clockwise and 1 counter clockwise)		Upon landing, the landing leg shoots out a metallic ballistic designed to penetrate Psyche's regolith and reach the metallic core. There would be a strong tether/chain attaching the ballistic to the lander. Upon hitting the metallic core of Psyche, a thermite payload would be ignited to weld the metallic ballistic to the Psyche's metal core.
		As we land, the legs will have suspension springs inside that will be self locking as they move down. The legs lock in order to remove any of the kinetic energy of the springs to launch the spacecraft back upwards		The legs will have two large segments with a hinge that will absorb the force of the impact. Since the gravity is ~1% Earth's gravity the total force experienced should be small.		Using the right surface location, we can design our spacecraft to wedge into a crater or canyon. the sides of the crater could also have loose debris that we could us to 'bury' the landing gear slightly to hold the spacecraft to the surface		By using an explosive device on the surface of Psyche could remove unnecessary debris that could make it impossible for us to drill/ weld/ etc into the surface to maintain surface connection. This debris that is blown can also be netted up to gather small samples of surface material. The man made crater could also be used to as a safer landing area.
		Two-stage attachment probe that first magnetically attaches to astroid surface with three legs, then has a second stage of three legs that have heated suction cups attach to the astroid and spary adhesive to attach. The probe would then drill down from the center of it below into the astroid		A large spinning drill launched from the orbitor down into the astroid, with a large platform deploying and stopping the drill once it makes contact with the surface. The teather will stay attached to the platfrom		The three legs of the probe all have adhesive where they make contact with the astroid, then heating to such a point that it melts into the iron. A stake is then attached to complete the anchor		Each foot of the anchor will actively drill into the asroid, with a stake inserted from above feet into the drilled holes

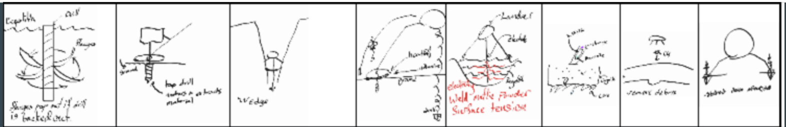
Pugh Chart/Down Select



Design Criterion	Weight (Scale 1-5: 1 = Low Importance, 5 = High Importance)	Desgin Concpts							
Lightweight	3	1	0	1	0	0	1	1	0
Cheap to manufacture	1	1	-1	1	-1	0	1	1	-1
Ability to anchor in near-zero gravity environments	5	1	-1	-1	1	0	1	0	1
Ability to anchor with an uneven surface	4	0	1	0	1	0	-1	-1	1
Ability to anchor into both a Regolith & Iron-Rich surface	3	1	1	-1	0	0	-1	-1	1
Concept Feasability	3	1	1	1	0	1	1	1	0
Compatibility with other capstone team solutions	5	1	1	1	0	1	-1	0	1
Attachment Speed	2	1	1	0	0	1	1	1	1
Successful attachment probability	4	0	-1	0	1	0	1	-1	1
Anchoring system must be able to live for long time without use (>10 year)	4	0	0	1	1	0	1	1	1
Able to withstand abrasive debris	1	1	-1	1	0	1	1	1	1
Total:	35	Summary/ Results							
Total +1		8	5	6	4	4	8	6	8
Total -1		0	4	2	1	0	3	3	1
Total 0		11	10	11	14	15	8	10	10
Weight Average		65.71%	17.14%	25.71%	45.71%	31.43%	31.43%	8.57%	77.14%

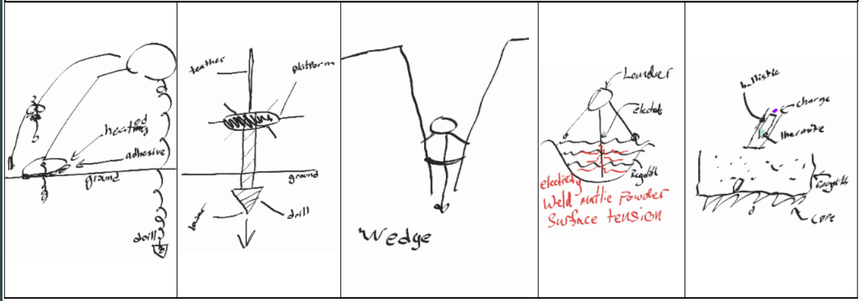
Instructions for Pugh Chart Use In Concept Generation-Refinement

When down-selecting concepts generated in the **Morph-Matrix** we need to consider various **Design Criterion** in which to analyze our concepts for further refinement. We have given a weighted score of importance ranked from 1-5 for each design criterion. The design concept generated from the **Morph-Matrix** are scored (1 / 0 / -1) based on (**Highly Meets/ Meets/ Does not Meet**) the Design Criteria and a **Weighted Average** is generated to provide a Percentage of How well the design Satisfies the overall Design Parameters. A weighted average of 100% implies that the associated design concept is best suited for the parameters of the project, scores with a 70% or greater are considered possible design approaches. While anything less than 70% should be rejected from consideration.(Subject to Change)



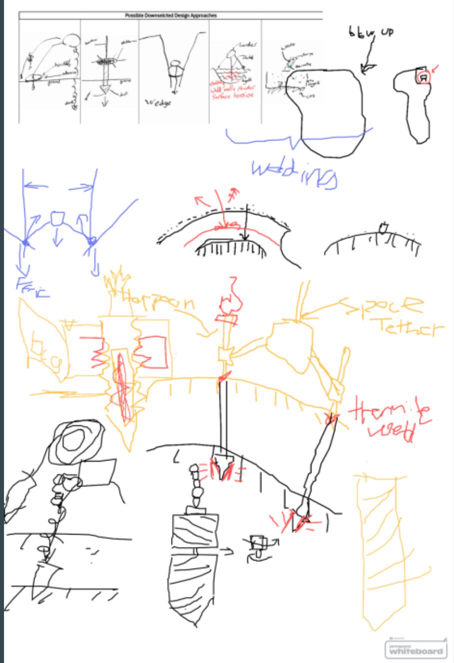
1	1	1	1	1	1	0	0
0	1	1	0	1	1	1	1
1	1	1	1	1	1	0	1
1	1	1	1	1	1	1	0
0	-1	0	0	0	1	0	0
1	1	1	1	1	1	1	1
1	1	1	1	1	1	0	0
0	-1	1	1	1	1	1	1
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1
6	8	9	8	9	10	6	6
0	2	0	0	0	0	0	0
13	9	10	11	10	9	13	13
68.57%	60.00%	80.00%	77.14%	80.00%	88.57%	42.86%	45.71%

Possible Downselcted Design Approaches

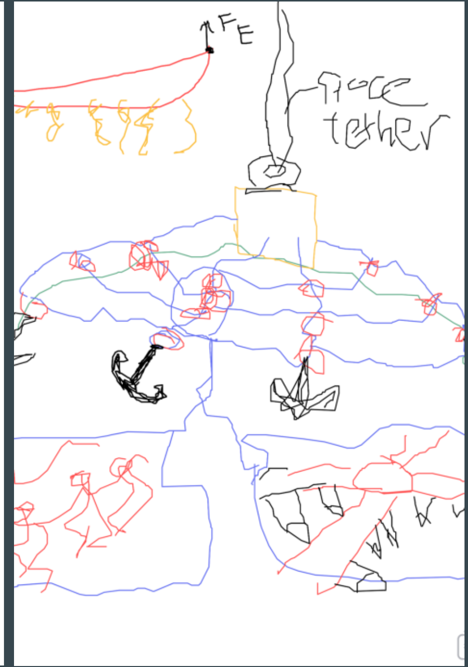


Each of these designs meets our requirement of at least 70% of the weighted design criteria per approach, and will each be used to create a design that is 100% of our required criteria. They were graded on 11 metrics, with the highest weights of their ability to anchor in near-zero gravity environments & their compatibility with other capstone team solutions. From this selection, we plan to choose based on a team-specific set of criteria based on ability to CAD, a deeper look into concept feasability, induvidual team skill matrices, and sponser input & rankings.

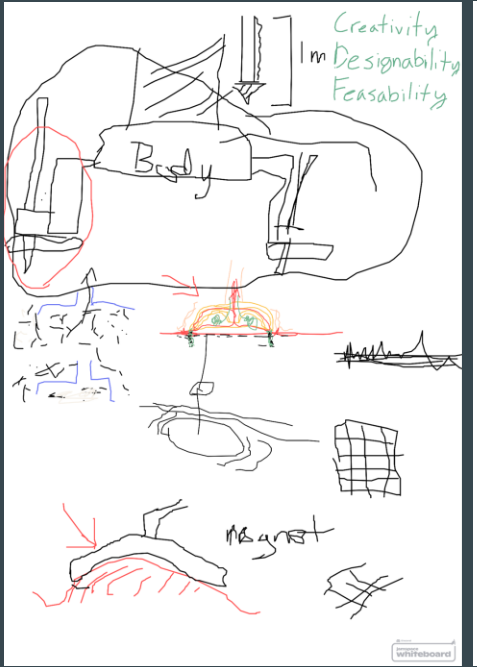
Down Select Concept Expansion



Regolith penetration ideas (ended up assuming no-regolith scenario)



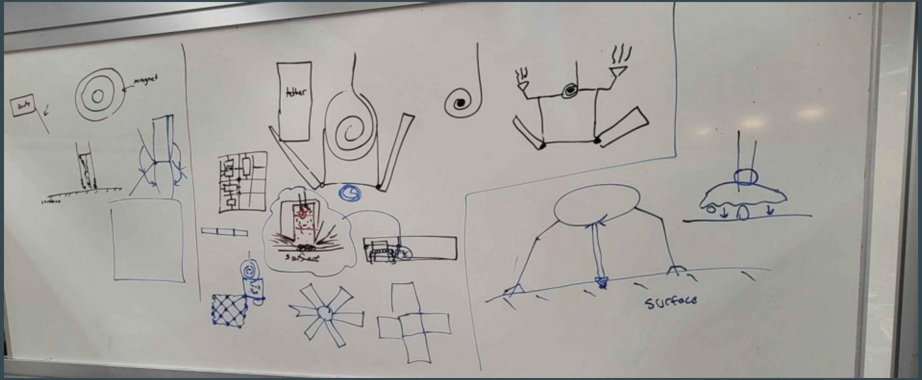
Netting based solution



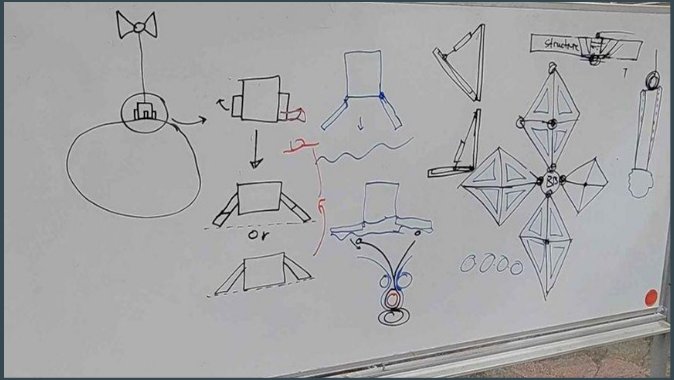
Drills, suction cups, and vibrational silt digger

Unfolding setup, thermite welding guided and initially anchored magnetically

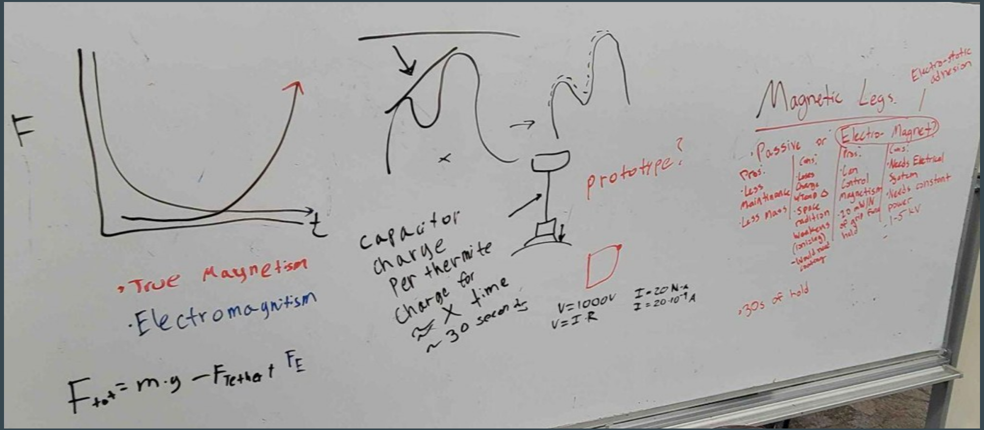
Library Concept Generation



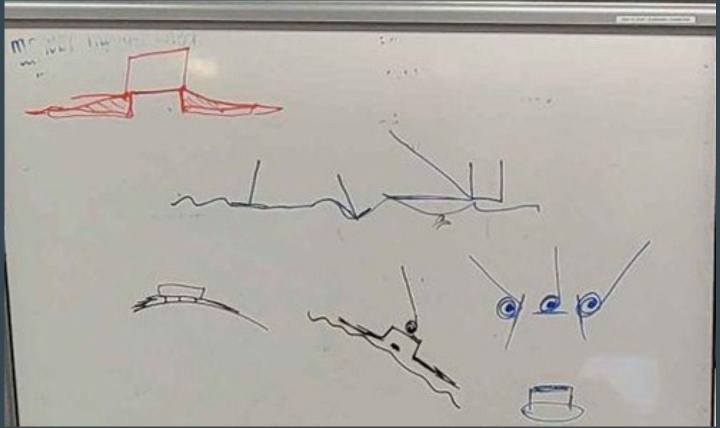
Different distributions of thermite charges and magnets



Folding leg designs

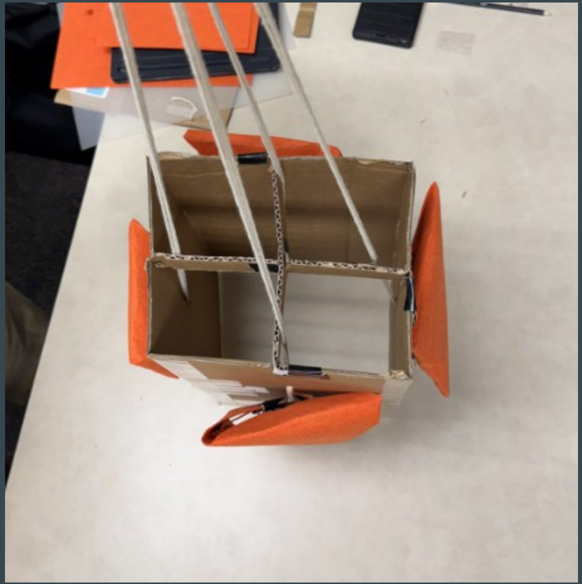


Magnetic options and forces on uneven surfaces

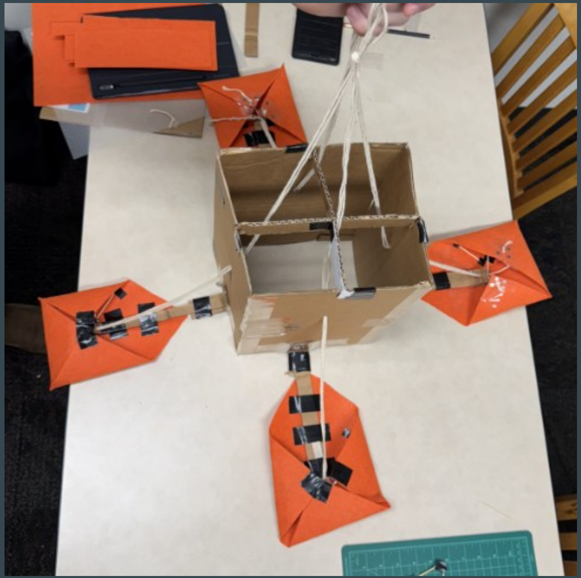


Tether gimbels for attachment to uneven surfaces

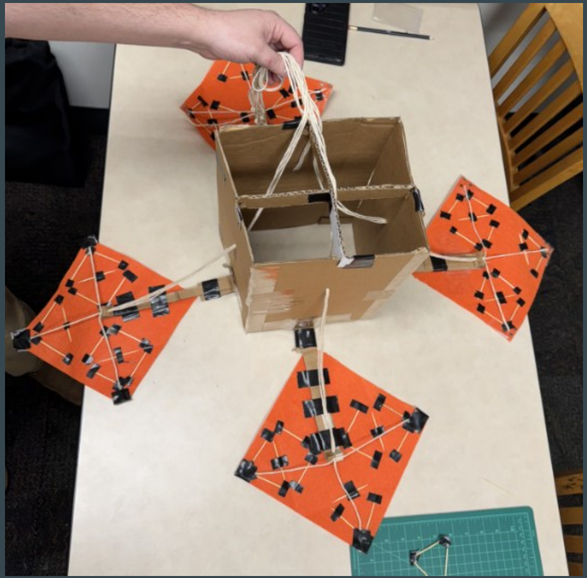
Prototype System Progression



Configuration during transit and decent



Configuration just above surface



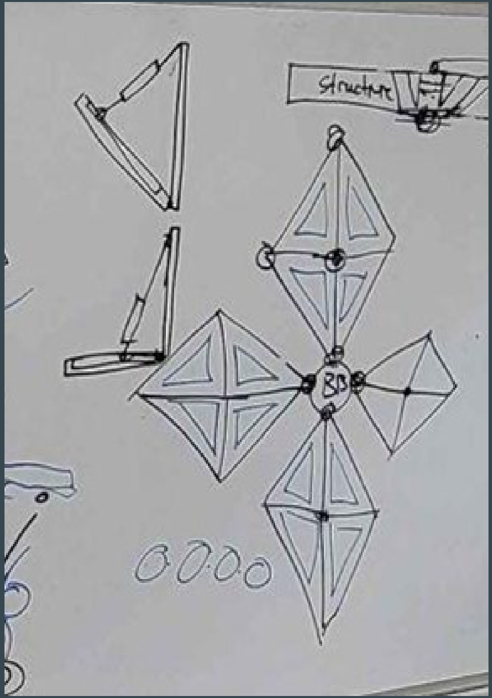
Configuration at surface

Physical Prototype

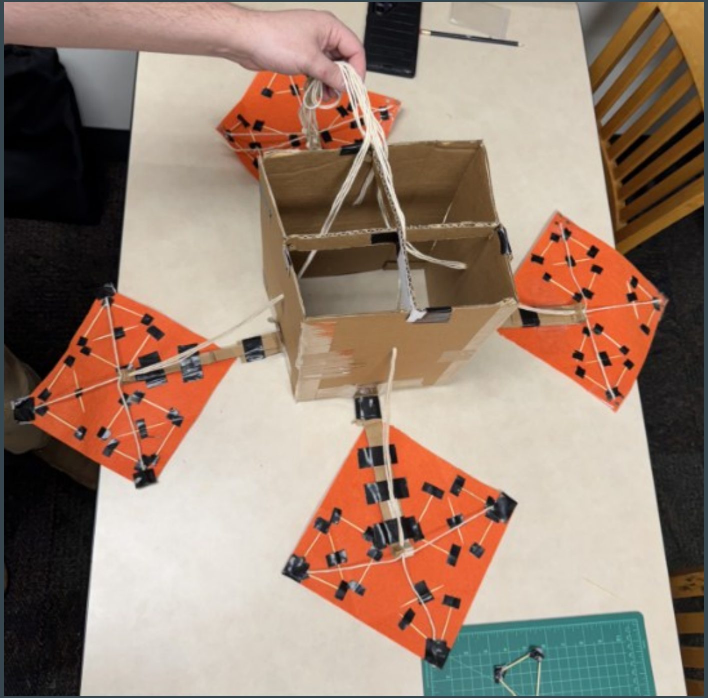
Worst case anchoring scenario in Earth gravity at 90 degrees

- Electroadhesion simulated with magnets.
- Attachment pads morph to environment.

Comparison of Design and Prototype Side-to-Side



Main leg design from library concept generation session



Prototype final configuration

Summary/Key Takeaways

- Electrostatic Adhesion
  - Proved viable for surface attachment
- Flexible Leg Design
  - Successfully adapted to uneven surface
- Prototype Performance
  - Demonstrated feasibility, with room for optimization.

Conclusion

- **Project Success**
  - Our prototyping phase validated key design principles, particularly in magnetic adhesion and structural adaptability.
- **Challenges & Improvements**
  - Some elements require further refinement, including material selection and magnetic efficiency.
- **Next Steps:**
  - Move towards advanced testing
  - Iterative design improvements (durability, electrostatic force efficiency, thermite integration)
  - Integration with a more robust control system

Thank you.